Chapter 5 & 7
Counting

MAD101

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Example

- How many 6-letter passwords ****** are there (the letter can be choose from 0..9;a..z;A..Z)?
- How many bit strings of length n that do not have two consecutive 0s?
- ......

The product rule

How many ways to choose the cloths?

The product rule

Definition 1.1

Task = task 1 $\rightarrow$ task 2 $\rightarrow \ldots \rightarrow$ task k

- Task 1: n_1 ways,
- Task 2: n_2 ways,
- ....
- Task k: n_k way.

Product rule: $n_1.n_2 \ldots n_k$ ways to do the task.

Example 1.1. From city A to city B there are 2 paths, from B to C there are 3 paths.

a. How many ways are there from A through B to C?

b. If two more paths are opened from A to C (without going through B), how many ways are there from A to C?

The sum rule

Definition 1.2

Task = task 1 or task 2 or....or task k

Task 1: n_1 ways,

Task 2: n_2 ways,

....

Task k: n_k way.

Sum rule: $n_1 + n_2 + ...n_k$ ways to do task.

Example 1.2. There are 37 faculties, 83 students. A person either a faculty or student can be choose to attend a committee. How many ways are there to select such person?

The sum rule

Definition 1.3

Task = task 1 or task 2 or....or task k

Task 1: n_1 ways,

Task 2: n_2 ways,

....

Task k: n_k way.

Sum rule: $n_1 + n_2 + ...n_k$ ways to do task.

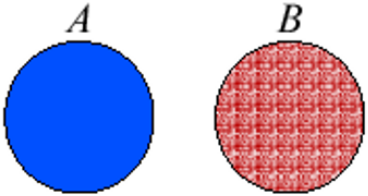
Example 1.3. There are 37 faculties, 83 students. A person either a faculty or student can be choose to attend a committee. How many ways are there to select such person?

Solution.

- First task (facalty): 37 ways
- Second task (student): 83 ways

Thus, Sum rule $\to 37 + 83 = 120$ ways.

Phrased in terms of Set Theory



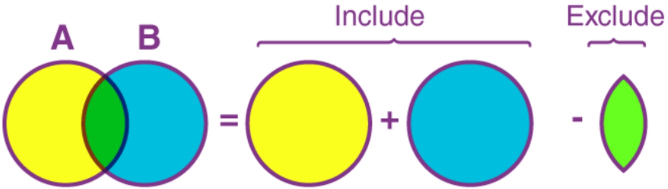
The sum and product rules can also be phrased in terms of Set Theory

- Sum rule: $|A \cup B| = |A| + |B|$
- Product rule: $|AB| = |A| \cdot |B|$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example 1.4. How many 8-bit strings either start with 1 or end with 00?

The Inclusion-Exclusion Principle



$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example 1.5. How many 8-bit strings either start with 1 or end with 00?

Solution.

- Start with 1: 2^7
- End with 00: 2^6
- Start with 1 **and** end with 00: 2^5

Thus, we have $2^7 + 2^6 - 2^5 = 160$ 8-bit strings.

Example

Example 1.6. How many different bit strings of length seven are there?

Example 1.7. Each user on a computer system has a password, which has properties:

1. six to eight characters long.
2. character is an uppercase letter or a digit.
3. contain at least one digit.

How many possible passwords are there?

Example 1.10. How many different bit strings of length seven are there?

Example 1.11. Each user on a computer system has a password, which has properties:

1. six to eight characters long.
2. character is an uppercase letter or a digit.
3. contain at least one digit.

How many possible passwords are there?

Solution. $(36^6 - 26^6) + (36^7 - 26^7) + (36^8 - 26^8)$.

Example 1.12. How many functions are there from $\{a, b, c\}$ to $\{1, 2, 3, 4, 5\}$?

Example 1.13. How many one-to-one functions are there from an 3-element set A to an 5-element set B?

Example

1 The basics of Counting

Example 1.14: Counting functions. How many functions are there from a set with m elements to a set with n elements?

Example

Example 1.16: Counting functions. How many functions are there from a set with m elements to a set with n elements?

Solution.

There are $n \cdot n \cdot \ldots \cdot n = n^m$ functions.

Example 1.17: Counting One-to-One Functions. How many one-to-one functions are there from a set with m elements to one with n elements?

Example 1.18: Counting functions. How many functions are there from a set with m elements to a set with n elements?

Solution.
There are $n \cdot n \cdot \ldots \cdot n = n^m$ functions.

Example 1.19: Counting One-to-One Functions. How many one-to-one functions are there from a set with m elements to one with n elements?

Solution.
Case 1: $m > n$ there are no one-to-one functions.
Case 2: $m \leq n$ there are $n(n-1)(n-2)\ldots(n-m+1)$ one-to-one functions.

Example 1.20. If there are 5 multiple-choice questions on an exam, each having four possible answers, how many different sequences of answers are there?

Example 1.22. If there are 5 multiple-choice questions on an exam, each having four possible answers, how many different sequences of answers are there?

Solution. $4^5 = 1024$.

Example 1.23. In how many ways can a teacher seat 5 girls and 3 boys in a row seats if a boy must be seated in the first and a girl in the last seat?

Example 1.24. If there are 5 multiple-choice questions on an exam, each having four possible answers, how many different sequences of answers are there?

Solution. $4^5 = 1024$.

Example 1.25. In how many ways can a teacher seat 5 girls and 3 boys in a row seats if a boy must be seated in the first and a girl in the last seat?

Solution.
Choose a boy for the first seat: 3 choices.
Choose a girl for the last seat: 5 choices.
Seat the remaining 6 students (2 boys + 4 girls) in the middle 6 seats: 6! ways.
$\Longrightarrow 3 \cdot 5 \cdot 6! = 10800$.

Example

Remark 1.1. If $n = p_1^{a_1} \cdot p_2^{a_2} \dots p_k^{a_k}$. Then the number of positive divisors of n is:

$$(a_1 + 1)(a_2 + 1) \dots (a_k + 1).$$

Example 1.26. How many positive divisors does 120 have?

Remark 1.2. If $n = p_1^{a_1} \cdot p_2^{a_2} \dots p_k^{a_k}$. Then the number of positive divisors of n is:

$$(a_1 + 1)(a_2 + 1) \dots (a_k + 1).$$

Example 1.30. How many positive divisors does 120 have?

Solution.
Prime factorization of 120: $120 = 2^3 \cdot 3^1 \cdot 5^1. \Longrightarrow (3 + 1)(1 + 1)(1 + 1) = 16.$

Example 1.31. $A = \{1, 2, 3, 4, 5, 6\}$

1. How many subsets of A can be constructed?
2. How many subsets of A that contain 1?
3. How many subsets neither contain 3 nor 4?

Example 1.32. How many integers between 1000 and 3000 inclusive are divisible by 3 or 7?

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2 Recurrence relations

Definition 2.1

A **recurrence relation** for the sequence $\{a_n\}$ is an equations that express a_n in terms of the previous terms $a_0, a_1, \ldots, a_{n-1}$.
The sequence $\{a_n\}$ is called a **solution** of the recurrence relation.

Example 2.1. Suppose $a_n = 2a_{n-1} - a_{n-2}$

- $a_n = 3n$ is a solution because of
  $2a_{n-1} - a_{n-2} = 2.3(n-1) - 3(n-2) = 3n = a_n$.
- $a_n = 5$ is a solution because of $2a_{n-1} - a_{n-2} = 2.5 - 5 = 5 = a_n$.
- $a_n = 2^n$ is not a solution because of
  $2a_{n-1} - a_{n-2} = 2.2^{n-1} - 2^{n-2} = 2^n - 2^{n-2} \neq a_n$.

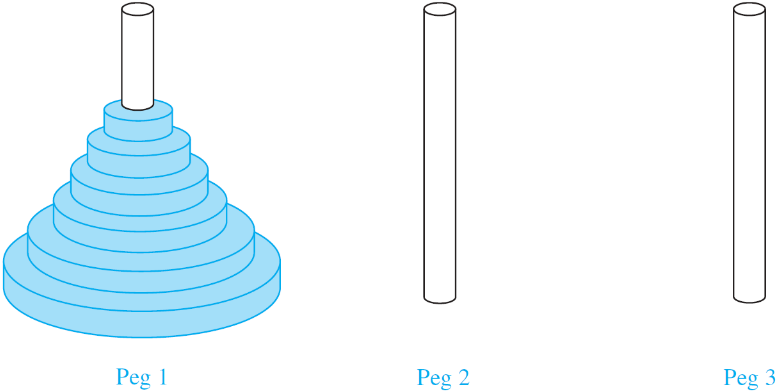
Remark 2.1. A recurrence relation may have more than one Solution.

Example

Find the solution of $a_n = 2a_{n-1} - a_{n-2}$, $\underbrace{a_0 = 3, a_1 = 5}_{\textbf{initial conditions}}$.

- $a_2 = 2a_1 - a_0 = 2.5 - 3 = 7 = 3 + 2.2$
- $a_3 = 2a_2 - a_1 = 2.7 - 5 = 9 = 3 + 2.3$
- $a_4 = 2a_3 - a_2 = 2.9 - 7 = 11 = 3 + 2.4$
- ....
- $a_n = 2a_{n-1} - a_{n-2} = 3 + 2.n$ is a solution.

The Tower of Hanoi Puzzle



How can we move the disks to the 3^{nd} peg, one in a time, following the rule: **larger disks are never placed on top of smaller ones**?

The Tower of Hanoi Puzzle - Solution

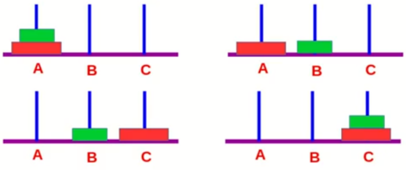
Let H_n denote the number of moves needed to solve the Tower of Hanoi puzzle with n disks. Set up a recurrence relation for the sequence $\{H_n\}$

- $n = 1$

Thus, $H_1 = 1$

The Tower of Hanoi Puzzle - Solution

- $n = 2$



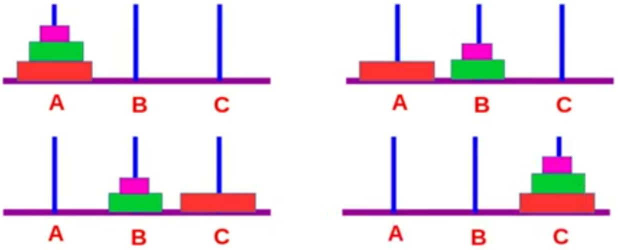
Move $1(2-1)$ disk from A to B.
Move 1 from A to C.
Move $1(2-1)$ from B to C.

Thus, $H_2 = 3$

The Tower of Hanoi Puzzle - Solution

2 Recurrence relations

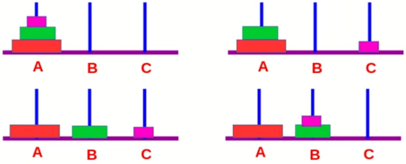
- $n = 3$



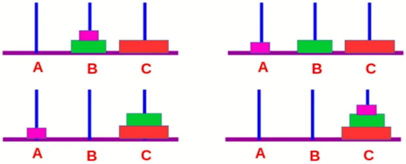
Move $2(3-1)$ disk from A to B.(?!)
Move 1 from A to C.
Move $2(3-1)$ from B to C.(?!)

The Tower of Hanoi Puzzle - Solution

Move $2(3-1)$ disk from A to B



Move $2(3-1)$ from B to C



Therefore, $H_3 = 7$

In General

We need to move n disks from A to C

Move $n-1$ disks from A to B.

Move 1 disk from A to C

Move $n-1$ disks from B to C

Thus, we have $H_1 = 1, H_n = 2H_{n-1} + 1$ is a recurrence relation.
To solve it, we write $H_n = 2H_{n-1} + 1 = 2[2H_{n-2} + 1] + 1 = 2^2 H_{n-2} + 2 + 1$
$= ... = 2^{n-1}H_1 + 2^{n-2} + ... + 2 + 1$
$H_1 = 1 \implies H_n = 2^n - 1$
Demo: https://mathsisfun.com/games/towerofhanoi.html

Remark 2.2. $\sum_{k=0}^{n} ar^k = a\dfrac{1-r^{n+1}}{1-r} (r \neq 1).$

Procedure hanoi(n,p1,p2,p3)

If $n = 1$ then

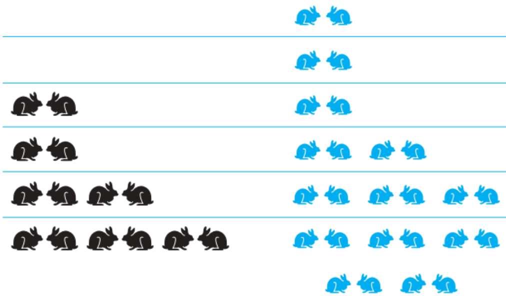
Print ($p1$,"$\rightarrow$",$p2$)

else

hanoi$(n-1, p1, p3, p2)$
hanoi$(1, p1, p2, p3)$
hanoi$(n-1, p3, p2, p1)$

Rabbits and the Fibonacci Numbers

A young pair of rabbits (1 male + 1 fem) is placed on an island. A 2 month old – pair will produce another pair. How many pairs of rabbits after n months?



Reproducing pairs (at least two months old)	Young pairs (less than two months old)	Month	Reproducing pairs	Young pairs	Total pairs
		1	0	1	1
		2	0	1	1
		3	1	1	2
		4	1	2	3
		5	2	3	5
		6	3	5	8

Rabbits and the Fibonacci Numbers - Solution

The number of pairs after the first few months are: $1, 1, 2, 3, 5, \ldots$

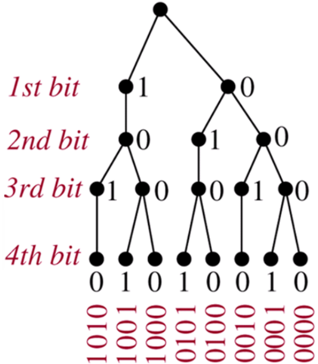
Let f_n be the number of pairs of rabbits after n months, the above results suggest the following relation:

$$f_n = f_{n-1} + f_{n-2}$$

- With the initial conditions $f_1 = 1$ and $f_2 = 1$ the solution is unique
- The numbers fn with the above initial conditions are also called the **Fibonacci numbers.**

Use recurrence relations for counting

Find recurrence relation and give initial conditions for **the number of bit strings of length n that do not have two consecutive 1s**. How many such bit strings are there of length four.



Solution

Let a_n: number of bit string of length n that do not have two consecutive 0s.

n	all 2^n bit strings of length n	a_n
1	0, 1	$a_1 = 2$
2	~~00~~, 01, 10, 11	$a_2 = 3$
3	~~000~~, ~~001~~, 010, 011, ~~100~~, 101, 110, 111	$a_3 = 5$
4	~~0000~~, ~~0001~~, ~~0010~~, ~~0011~~, ~~0100~~, 0101, 0110, 0111, ~~1000~~, ~~1001~~, 1010, 1011, ~~1100~~, 1101, 1110, 1111	$a_4 = 8$
…	…	…

$$\Longrightarrow a_n = \begin{cases} 2, & n = 1 \\ 3, & n = 2 \\ a_{n-1} + a_{n-2}, & n \geq 3 \end{cases}$$

Tree diagram

Counting problems can be solved using tree diagrams.

Example 2.2. Suppose that "I Love New Jersey" T-shirts come in five different sizes: S, M, L, XL, and XXL. Further suppose that each size comes in four colors, white, red, green, and black, except for XL, which comes only in red, green, and black, and XXL, which comes only in green and black. How many different shirts does a souvenir shop have to stock to have at least one of each available size and color of the T-shirt?

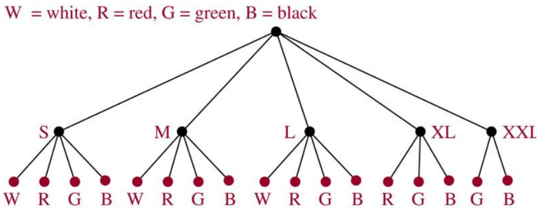
Tree diagram

2 Recurrence relations

Counting problems can be solved using tree diagrams.

Example 2.3. Suppose that "I Love New Jersey" T-shirts come in five different sizes: S, M, L, XL, and XXL. Further suppose that each size comes in four colors, white, red, green, and black, except for XL, which comes only in red, green, and black, and XXL, which comes only in green and black. How many different shirts does a souvenir shop have to stock to have at least one of each available size and color of the T-shirt?

Solution.



Compound interest Problem (lãi kép)

A sum of P_0 is deposited in the saving account with the interest rate of $r\%$ compounded annually. How much will be in the account after n years?
Let P_n be the amount after n years, then P_n is obtained from P_{n-1} and the interest yielded from that amount in one year:

$$P_n = P_{n-1} + rP_{n-1} = (1+r)P_{n-1} = (1+r)^2 P_{n-2} = ... = (1+r)^n P_0$$

Thus, $P_n = (1+r)^n P_0$

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3 Divide & Conquer Algorithms

Example 3.1. $3, 5, 6, 8, 9, 11, 12, 15, 16, 19, 20, 25$. Use Binary Search, find 12.

L					M						R
0	1	2	3	4	5	6	7	8	9	10	11
3	5	6	8	9	10	12	15	16	19	20	25

Example 3.2. 3, 5, 6, 8, 9, 11, 12, 15, 16, 19, 20, 25. Use Binary Search, find 12.

L					M						R
0	1	2	3	4	5	6	7	8	9	10	11
3	5	6	8	9	10	12	15	16	19	20	25

						L		M			R
0	1	2	3	4	5	6	7	8	9	10	11
3	5	6	8	9	10	12	15	16	19	20	25

Example 3.3. 3, 5, 6, 8, 9, 11, 12, 15, 16, 19, 20, 25. Use Binary Search, find 12.

L					M						R
0	1	2	3	4	5	6	7	8	9	10	11
3	5	6	8	9	10	12	15	16	19	20	25

						L		M			R
0	1	2	3	4	5	6	7	8	9	10	11
3	5	6	8	9	10	12	15	16	19	20	25

						L-M	R				
0	1	2	3	4	5	6	7	8	9	10	11
3	5	6	8	9	10	12	15	16	19	20	25

This is a Divide-and-Conquer Algorithms.

Divide-and-Conquer Algorithms

- To solve a problem of size n,
- (Divide.) We divide the original problem into subproblems that are similar to the original problem but of size smaller than n,
- (Conquer.) Solving the subproblems and summarizing the solutions, we get the solution of the original problem,
- For subproblems we also apply the same approach,
- This division gives us basic problems that are easy to solve,
- $\rightarrow$ Recurrence relations algorithms.

Divide-and-conquer recurrence relation

- n: size of the original problem,
- a: numbers of sub-problems must be conquered,
- n/b : size of the sub-problem,
- $f(n)$: number of operation required to solve the original problem,
- $\rightarrow f(n/b)$: number of operation required to solve a sub-problem,
- $g(n)$: overhead for additional work of the step conquer.

Divide-and-conquer recurrence relation:

$$\boxed{f(n) = af\left(\frac{n}{b}\right) + g(n)}$$

Theorem 3.1

Let f be an increasing function that satisfies the recurrence relation $f(n) = af(n/b) + c$ whenever n is divisible by b, where $a \geq 1$, b is an integer greater than 1, and c is a positive real number. Then

$$f(n) = \begin{cases} O(n^{\log_b a}), & \text{if } a > 1, \\ O(\log n), & \text{if } a = 1. \end{cases}$$

Furthermore, when $n = b^k$ and $a \neq 1$, where k is a positive integer,

$$f(n) = C_1 n^{\log_b a} + C_2,$$

where $C_1 = f(1) + \dfrac{c}{a-1}$ and $C_2 = \dfrac{-c}{a-1}$.

Remark 3.1. $n = b^k \implies n^{log_b a} = (b^k)^{log_b a} = b^{log_b a^k} = a^k$. Hence,

$$f(n) = C_1 a^k + C_2 = a^k[f(1) + c/(a-1)] - c/(a-1)$$

Example 3.4. Estimate the number of comparisons used by a binary search

$$f(n) = f(n/2) + 2$$

Example 3.5. Estimate the number of comparisons to locate the maximum element in a sequence $f(n) = 2f(n/2) + 1$.

Example 3.6. $f(n) = 5f(n/2) + 3, f(1) = 7$. Find $f(2^k), k$ is a positive integer. Estimate $f(n)$ if f is increasing function.

Example 3.7. Estimate the number of comparisons used by a binary search

$$f(n) = f(n/2) + 2$$

Example 3.8. Estimate the number of comparisons to locate the maximum element in a sequence $f(n) = 2f(n/2) + 1$.

Example 3.9. $f(n) = 5f(n/2) + 3, f(1) = 7$. Find $f(2^k), k$ is a positive integer. Estimate $f(n)$ if f is increasing function.

Solution.

$f(2^k) = 5^k[f(1) + c/(a-1)] - c/(a-1) = 5^k[7 + 3/4] - 3/4 = \dfrac{31}{4}.5^k - \dfrac{3}{4}$

$a = 5 > 1$, f increasing, so $f(n) = O(n^{log5})$

Master Theorem

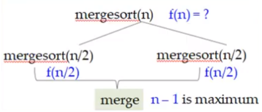
Theorem 3.2: MASTER THEOREM

Let f be an increasing function that satisfies the recurrence relation

$$f(n) = af(n/b) + cn^d$$

whenever $n = b^k$, where k is a positive integer, $a \geq 1$, b is an integer greater than 1, and c and d are real numbers with c positive and d nonnegative. Then

$$f(n) = \begin{cases} O(n^d) & \text{if } a < b^d, \\ O(n^d \log n) & \text{if } a = b^d, \\ O(n^{\log_b a}) & \text{if } a > b^d. \end{cases}$$



- $f(n) = 2f(n/2) + n \rightarrow$
  $a = 2, b = 2, d = 1, c = 1 \rightarrow a = b^d$
- $\rightarrow f(n)$ is $O(nlogn)$

Example

Example 3.10. Complexity of Merge Sort

$$M(n) = 2M(\frac{n}{2}) + n$$

Example 3.11. Give a big-O estimate for the number of bit operations needed to multiply two n-bit integers using the fast multiplication algorithm

$$f(n) = 3f\left(\frac{n}{2}\right) + Cn$$

Example 3.12. Give a big-O estimate for the number of multiplications and additions required to multiply two nxn matrices using the matrix multiplication algorithm $f(n) = 7f\left(\frac{n}{2}\right) + 15\frac{n^2}{4}$, when n is even.

Q&A

Thank you for listening!